

Linear Equations in Two Variables

Chapter 3 - Hinglish Notes, Methods, Examples + Solutions

Ye Chapter 3 (Pair of Linear Equations in Two Variables) ka complete guide hai. Pehle red-page ke core basics (jo Class 7-8-9 me padhe), fir chapter ka main content - graphical, substitution aur elimination methods + a1/a2 conditions. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. Variable aur Equation

EXAM * Exam me CONSTANTLY aata hai

Variable: Unknown number, jaise x ya y .

Equation: Aisa statement jisme '=' (barabar) ka chihni ho.

- Example: $2x + 3 = 7$ ek equation hai.

2. Linear Equation in ONE Variable

EXAM * Exam me CONSTANTLY aata hai

Form: $ax + b = 0$ (sirf ek variable, power 1).

- Example: $3x - 6 = 0 \rightarrow x = 2$.
- Ise sirf EK hi solution milta hai.

3. Linear Equation in TWO Variables

EXAM * Exam me CONSTANTLY aata hai

Form: $ax + by + c = 0$ (do variables, dono ki power 1).

- Example: $2x + 3y = 12$.
- Iske INFINITE (anant) solutions hote hain - har x ke liye ek y .

4. Solution Kya Hai?

EXAM * Exam me CONSTANTLY aata hai

Definition: Wo (x, y) jodi jo equation ko sahi (satisfy) kare.

- Example: $x + y = 5$ ke liye $(2, 3)$ solution hai, kyunki $2 + 3 = 5$.

5. Graph Basics

EXAM * Exam me CONSTANTLY aata hai

- X-axis = horizontal (leti), Y-axis = vertical (khadi).
- Dono ORIGIN $(0, 0)$ par milte hain. Point ko (x, y) likhte hain.
- Linear equation ka graph hamesha ek SEEDHI LINE hoti hai.

6. Ratio (a1/a2 type)

Ratio: Do numbers ki tulna, jaise $2/4 = 1/2$.

- Is chapter me coefficients ke ratio compare karenge: a_1/a_2 , b_1/b_2 , c_1/c_2 .

7. Simple Equation Solve Karna

EXAM * Exam me CONSTANTLY aata hai

Transposition Ek side ki cheez doosri side le jao to uska sign badal jata hai.

- Example: $2x = 18 \rightarrow x = 18/2 = 9$.
- Jo jod raha tha wo ghatne lagta hai; jo guna kar raha tha wo bhaag.

PART B - Chapter 3 ka Main Content

8. Pair of Linear Equations

Matlab: Do linear equations jo ek saath (simultaneously) di gayi hon, jaise:

- $a_1x + b_1y + c_1 = 0$
- $a_2x + b_2y + c_2 = 0$
- Inhe ek saath solve karke wo (x, y) dhoondte hain jo DONO ko satisfy kare.

9. Graphical Method (3 Cases)

EXAM * Exam me CONSTANTLY aata hai

- Dono equations ki seedhi lines banao aur dekho wo kaise milti hain:
- Case 1: Lines ek point par CUT karein $\rightarrow$ 1 unique solution.
- Case 2: Lines PARALLEL hon $\rightarrow$ koi solution nahi (no solution).
- Case 3: Lines bilkul UPAR-UPAR (coincident) $\rightarrow$ infinite solutions.

10. Conditions for Solutions (a_1/a_2 Rule)

EXAM * Exam me CONSTANTLY aata hai

Ye sabse important exam point hai. Coefficients ke ratio se solution ka type pata chalta hai:

$a_1/a_2 \neq b_1/b_2 \rightarrow$ Unique solution (lines intersect) [Consistent]
 $a_1/a_2 = b_1/b_2 \neq c_1/c_2 \rightarrow$ No solution (parallel lines) [Inconsistent]
 $a_1/a_2 = b_1/b_2 = c_1/c_2 \rightarrow$ Infinite solutions (coincident) [Consistent]

- Consistent = solution hai; Inconsistent = koi solution nahi.

11. Substitution Method

EXAM * Exam me CONSTANTLY aata hai

- Step 1: Ek equation se ek variable ko doosre ke roop me likho (jaise $x = \dots$).
- Step 2: Use doosri equation me substitute (daal) karo $\rightarrow$ ek variable bachega.
- Step 3: Use solve karo, fir value wapas daal kar doosra variable nikalo.

12. Elimination Method

EXAM * Exam me CONSTANTLY aata hai

- Step 1: Coefficients barabar karne ke liye equations ko number se multiply karo.
- Step 2: Dono equations ko jodo ya ghatao taaki ek variable HAT (eliminate) jaye.
- Step 3: Bache hue variable ko solve karo, fir doosra nikalo.

13. Word Problems

- Sentences ko x aur y me likho (do unknowns -> do equations banao).
- Fir substitution ya elimination se solve karo.
- Common types: numbers ka sum/difference, age problems, cost of items.

Quick Revision

One variable -> 1 solution; Two variables -> infinite (akeli eq).

Solution = (x, y) jo equation satisfy kare.

$a_1/a_2 \neq b_1/b_2 \rightarrow$ unique; $a_1/a_2 \neq b_1/b_2 \neq c_1/c_2 \rightarrow$ none; all equal -> infinite.

Methods: Graphical, Substitution, Elimination.

Linear graph hamesha seedhi line.

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1 (Substitution): $x + y = 5$, $x - y = 1$

- Step: Eq1 se: $x = 5 - y$.
- Step: Eq2 me daalo: $(5 - y) - y = 1 \rightarrow 5 - 2y = 1$.
- Step: $2y = 4 \rightarrow y = 2$. Fir $x = 5 - 2 = 3$.

Answer: $x = 3$, $y = 2$

Example 2 (Elimination): $2x + 3y = 12$, $2x - y = 4$

- Step: Dono me 2x same hai, to ghatao (Eq1 - Eq2):
- Step: $(2x + 3y) - (2x - y) = 12 - 4 \rightarrow 4y = 8 \rightarrow y = 2$.
- Step: $2x - 2 = 4 \rightarrow 2x = 6 \rightarrow x = 3$.

Answer: $x = 3$, $y = 2$

Example 3 (Condition): $2x + 3y - 5 = 0$, $4x + 6y - 10 = 0$

- Step: $a_1/a_2 = 2/4 = 1/2$.
- Step: $b_1/b_2 = 3/6 = 1/2$.
- Step: $c_1/c_2 = -5/-10 = 1/2$. Teeno barabar.

Answer: Infinite solutions (coincident lines)

Example 4 (Condition): $x + 2y - 4 = 0$, $2x + 4y - 12 = 0$

- Step: $a_1/a_2 = 1/2$, $b_1/b_2 = 2/4 = 1/2$.
- Step: $c_1/c_2 = -4/-12 = 1/3$.
- Step: $a_1/a_2 = b_1/b_2$ but $\neq c_1/c_2$.

Answer: No solution (parallel lines)

Example 5 (Condition): $x + y - 5 = 0$, $x - y - 1 = 0$

- Step: $a_1/a_2 = 1/1 = 1$.
- Step: $b_1/b_2 = 1/(-1) = -1$.
- Step: $1 \neq -1$, to $a_1/a_2 \neq b_1/b_2$.

Answer: Unique solution (intersecting)

Example 6 (One variable): $3x - 6 = 0$

- Step: $3x = 6$.
- Step: $x = 6/3$.

Answer: $x = 2$

Example 7 (Transposition): $2x = 18$

- Step: $x = 18 / 2$.

Answer: $x = 9$

Example 8 (Check solution): Kya (2, 3) $x + y = 5$ ka solution hai?

- Step: $x=2, y=3$ daalo: $2 + 3 = 5$.
- Step: Equation satisfy ho gayi.

Answer: Haan, solution hai

Example 9 (Word problem): Do numbers ka sum 10, difference 4.

- Step: Maano numbers x aur y : $x + y = 10$, $x - y = 4$.
- Step: Jodo: $2x = 14 \rightarrow x = 7$.
- Step: $y = 10 - 7 = 3$.

Answer: Numbers = 7 aur 3

Example 10 (Substitution): $y = 2x$, $x + y = 9$

- Step: $y = 2x$ ko doosri me daalo: $x + 2x = 9$.
- Step: $3x = 9 \rightarrow x = 3$. Fir $y = 2(3) = 6$.

Answer: $x = 3, y = 6$

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

Q1. Linear equation in two variables ka general form kya hai?

Q2. $ax + b = 0$ ke kitne solutions hote hain?

Q3. $2x + 3y = 12$ ke kitne solutions hote hain?

Q4. Kya (1, 4) equation $x + y = 5$ ka solution hai?

Q5. Solve karo: $x + y = 7$, $x - y = 3$.

Q6. Substitution se solve: $y = 2x$, $x + y = 9$.

Q7. $a_1/a_2 \neq b_1/b_2$ ho to kitne solutions?

Q8. $a_1/a_2 = b_1/b_2 \neq c_1/c_2$ ho to kya hota hai?

Q9. $a_1/a_2 = b_1/b_2 = c_1/c_2$ ho to kitne solutions?

Q10. Linear equation ka graph kaisa hota hai?

Q11. Origin ke coordinates kya hote hain?

Q12. Solve karo: $2x = 16$.

Q13. Parallel lines wale system ko consistent kehte hain ya inconsistent?

Q14. Coincident lines ke kitne solutions hote hain?

Q15. Elimination se solve: $x + y = 10$, $x - y = 2$.

Q16. $3x - 9 = 0$ ka solution kya hai?

Q17. Intersecting lines ke kitne solutions hote hain?

Q18. X-axis aur Y-axis kahan milte hain?

Q19. Do numbers ka sum 12 aur difference 2 hai. Numbers batao.

Q20. Solve karo: $5x = 35$.

Step-by-Step Solutions

Q1. General form?

- Step: Do variables, power 1, '=' ke saath.

Answer: $ax + by + c = 0$

Q2. $ax + b = 0$ ke solutions?

- Step: Ek variable, power 1 -> ek hi value milti hai.

Answer: 1 (unique)

Q3. $2x + 3y = 12$ ke solutions?

- Step: Two variables -> har x ke liye ek y.

Answer: Infinite

Q4. Kya (1, 4) $x + y = 5$ ka solution?

- Step: $x=1, y=4$: $1 + 4 = 5$. Satisfy.

Answer: Haan

Q5. $x + y = 7, x - y = 3$?

- Step: Jodo: $2x = 10$ -> $x = 5$.

- Step: $y = 7 - 5 = 2$.

Answer: $x=5, y=2$

Q6. $y = 2x, x + y = 9$?

- Step: $x + 2x = 9$ -> $3x = 9$ -> $x = 3$.

- Step: $y = 2(3) = 6$.

Answer: $x=3, y=6$

Q7. $a_1/a_2 \neq b_1/b_2$?

- Step: Lines intersect karti hain.

Answer: Unique solution

Q8. $a_1/a_2 = b_1/b_2 \neq c_1/c_2$?

- Step: Lines parallel hoti hain.

Answer: No solution (inconsistent)

Q9. $a_1/a_2 = b_1/b_2 = c_1/c_2$?

- Step: Lines coincident (upar-upar) hoti hain.

Answer: Infinite solutions

Q10. Linear equation ka graph?

- Step: Degree 1 -> seedhi line.

Answer: Straight line

Q11. Origin ke coordinates?

- Step: X aur Y axis jahan milte hain.

Answer: (0, 0)

Q12. $2x = 16$?

- Step: $x = 16/2$.

Answer: $x = 8$

Q13. Parallel lines -> consistent ya inconsistent?

- Step: Koi solution nahi hota.

Answer: Inconsistent

Q14. Coincident lines ke solutions?

- Step: Lines bilkul same hain.

Answer: Infinite

Q15. $x + y = 10$, $x - y = 2$?

- Step: Jodo: $2x = 12 \rightarrow x = 6$.
- Step: $y = 10 - 6 = 4$.

Answer: $x=6$, $y=4$

Q16. $3x - 9 = 0$?

- Step: $3x = 9 \rightarrow x = 9/3$.

Answer: $x = 3$

Q17. Intersecting lines ke solutions?

- Step: Ek hi point par milti hain.

Answer: 1 (unique)

Q18. X-axis aur Y-axis kahan milte?

- Step: Dono ka milan bindu = origin.

Answer: Origin (0,0)

Q19. Sum 12, difference 2?

- Step: $x + y = 12$, $x - y = 2$.
- Step: Jodo: $2x = 14 \rightarrow x = 7$, $y = 5$.

Answer: 7 aur 5

Q20. $5x = 35$?

- Step: $x = 35/5$.

Answer: $x = 7$